$$C = \overline{(1 \cdot 1)} = \overline{1} = 0$$

The truth table for NAND gate is

Input		Output
A	B	C
0	0	1
0	1	1
1	0	1
1	1	0

Table 2.11 Truth Table for NAND Gate

2.9.7 Bubbled OR Gate

The logic circuit of bubbled OR gate is

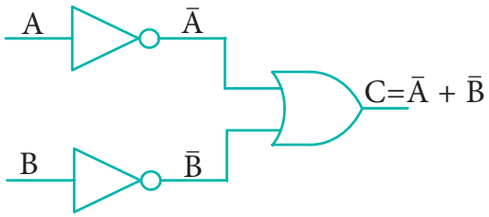


Fig. 2.13 Logic Circuit of Bubbled OR Gate

The output of this circuit can be written as $C = \overline{A} + \overline{B}$

The above circuit can be redrawn as the bubbles on the input, where the bubbles represents the inversion.

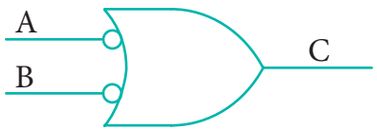


Fig. 2.14 Logic Symbol of Bubbled OR Gate

We refer this as bubbled OR gate. The truth table for the bubbled OR is

Input		Output
A	B	C
0	0	1
0	1	1
1	0	1
1	1	0

Table 2.12 Truth Table for Bubbled OR Gate

If we compare the truth tables of the bubbled OR gate with NAND gate, they are identical. So the circuits are interchangeable.

Therefore,

$$\overline{(A \cdot B)} = \overline{A} + \overline{B}$$

Which establishes the De Morgan's second theorem.

2.9.8 XOR Gate

The XOR (exclusive - OR) gate acts in the same way as the logical "either/or." The output is "true" if either, but not both, of the inputs are "true". The output is "false" if both inputs are "false" or if both inputs are "true." Another way of looking at this circuit is to observe that the output is 1 if the inputs are different, but 0 if the inputs are the same. The logic circuit of XOR gate is

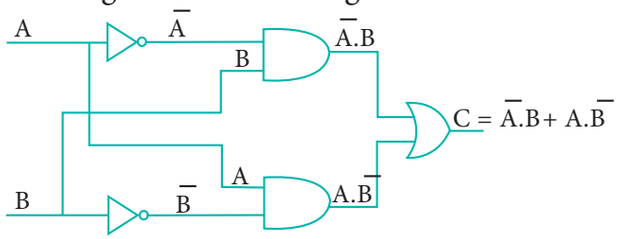


Fig. 2.15 Logic Circuit of XOR Gate

The output of the XOR gate is

The truth table for XOR gate is

Input		Output
A	B	C
0	0	0
0	1	1
1	0	1
1	1	0

Table 2.13 Truth Table for XOR Gate

In boolean algebra. exclusive - OR operator $\oplus$ or "encircled plus".

Hence $C = A \oplus B$

The logical symbol of XOR gate is

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